Does Optimization Beat 1/N? Evidence from an 11-ETF Multi-Asset Universe, 2012–2026

Investment Research Portfolio · Project 6 · Data to 30 September 2026

Abstract

DeMiguel, Garlappi and Uppal (2009) find that none of 14 optimizing models reliably beats an equally weighted (1/N) portfolio out of sample. I re-run their question on 11 liquid US-listed multi-asset ETFs with monthly rebalancing, a rolling 60-month estimation window, a 40% position cap and 10 basis points of one-way trading costs, over 173 out-of-sample months (May 2012 to September 2026). The classic 60/40 benchmark has the highest net Sharpe ratio (0.83 vs. 0.45 for 1/N; stationary-bootstrap p = 0.012). Contrary to the usual finding, the naive sample-moment maximum-Sharpe portfolio also beats 1/N after costs (Sharpe 0.80, difference +0.35, p = 0.034), but the result is fragile: it is insignificant with a 36-month window (p = 0.102), at 25 bp costs (p = 0.054) and in each subperiod on its own. Minimum variance, risk parity and Black-Litterman are statistically indistinguishable from 1/N. In this period 1/N was a weak benchmark: it held only 9% in SPY, which returned 14.5% a year (no other ETF exceeded 7.9%), and 36% in TLT, IEF, TIP, DBC, the four worst performers.

1. Introduction

Mean-variance optimization (Markowitz, 1952) is optimal only if expected returns and covariances are known. In practice they are estimated, and estimation error in expected returns is large enough that optimized portfolios often do worse out of sample than naive rules. DeMiguel, Garlappi and Uppal (2009, *Review of Financial Studies* 22(5), 1915–1953) showed that with realistic estimation windows the 1/N rule is hard to beat in Sharpe ratio, certainty-equivalent return or turnover. This study asks the same question for a practitioner's multi-asset universe: **does any of five standard construction methods beat a simple equal-weight portfolio after transaction costs?**

The six strategies span the main families used by multi-asset teams: a fixed benchmark (60/40), a risk-only optimizer (minimum variance with Ledoit-Wolf shrinkage), a full mean-variance optimizer with naive inputs (maximum Sharpe), a risk-budgeting rule (equal risk contribution) and a Bayesian blend of an equilibrium prior with a systematic view (Black-Litterman with a momentum view).

2. Data

The universe is 11 US-listed ETFs covering every major liquid asset class: SPY, EFA, EEM (equities), IEF, TLT, TIP (Treasuries and TIPS), LQD, HYG (credit), VNQ (REITs), GLD and DBC (real assets). Daily dividend- and split-adjusted closing prices come from Yahoo Finance via yfinance; I convert them to month-end total returns. HYG launched in April 2007, so the common sample runs from May 2007 to September 2026 (233 months). The risk-free rate is the 3-month Treasury bill (FRED series TB3MS), converted to a monthly rate. All inputs are saved in `/data`.

ETF	Ann. mean	Ann. vol	Worst month	Best month
SPY	11.5%	15.5%	-16.5%	12.7%
EFA	5.9%	17.3%	-20.8%	14.3%
EEM	6.9%	21.0%	-25.6%	16.9%
IEF	3.1%	6.7%	-4.7%	7.8%
TLT	3.4%	14.1%	-13.1%	14.3%
TIP	3.3%	5.8%	-8.1%	6.5%
LQD	4.0%	8.2%	-10.7%	13.8%
HYG	5.1%	10.2%	-11.5%	16.8%
VNQ	7.4%	22.2%	-31.7%	30.7%
GLD	10.5%	17.6%	-16.1%	12.8%
DBC	4.1%	19.1%	-25.1%	16.3%

Table 1. Monthly total returns of the 11 ETFs, May 2007 – September 2026 (arithmetic mean ×12, volatility ×√12).

3. Method

3.1 Backtest design

- Rolling estimation window of 60 months; weights for month t use only returns through month t–1. The first out-of-sample month is May 2012 and the last is September 2026.
- Monthly rebalancing to target weights; long-only; maximum 40% in any ETF.
- Trading costs of 10 bp one-way on turnover, where turnover is the sum of absolute differences between the new target and the drifted weights of the previous month. The initial purchase is not charged.

3.2 The six strategies

- **1/N:** 9.09% in each ETF, rebalanced monthly.
- **60/40:** 60% SPY, 40% IEF, rebalanced monthly.
- **Minimum variance:** minimizes portfolio variance using the Ledoit-Wolf (2004) shrunk covariance matrix (constant-variance target), solved with PyPortfolioOpt.
- **Maximum Sharpe:** tangency portfolio from sample means and the sample covariance matrix of the window, with the window's average T-bill rate as the risk-free rate. In every window at least one ETF had a mean return above cash, so the tangency portfolio was always defined.
- **Risk parity:** equal risk contribution weights from the sample covariance matrix, solved with the convex formulation of Spinu (2013). The 40% cap never bound.
- **Black-Litterman:** prior weights SPY 35%, EFA 15%, EEM 5%, IEF 15%, TLT 5%, TIP 5%, LQD 10%, HYG 5%, VNQ 2%, GLD 2%, DBC 1%; risk aversion $\delta = 2.5$; $\tau = 0.05$; implied excess returns $\pi = \delta\Sigma w$. Each month one relative view: the three ETFs with the highest 12-1 month momentum (months t–12 to t–2) outperform the three lowest by 3% a year, with 50% confidence using Idzorek's (2005) method. Final weights maximize $\mu'w - (\delta/2)w'\Sigma w$ on the posterior moments subject to the same long-only, 40% constraints.

3.3 Performance measures and test

I report the annualized geometric return, annualized volatility, Sharpe and Sortino ratios of monthly excess returns over TB3MS (annualized by √12), maximum drawdown, Calmar ratio (return ÷ |max drawdown|), average monthly turnover and the effective number of holdings, $1/\Sigma w^2$. To test whether a Sharpe ratio differs from that of 1/N, I resample the paired monthly excess returns with the stationary bootstrap of Politis and Romano (1994), 5,000 draws, mean block length chosen by the Politis-White (2004) rule. The two-sided p-value is the share of re-centred bootstrap differences at least as large in absolute value as the observed difference.

4. Results

Strategy	Ann. return	Vol.	Sharpe	Sortino	Max DD	Calmar	Turnover /mo	Eff. N	Δ Sharpe vs 1/N	p-value
1/N	5.2%	8.2%	0.45	0.66	-18.5%	0.28	2.1%	11.0	–	–
60/40	9.2%	9.0%	0.83	1.31	-20.5%	0.45	1.6%	1.9	+0.38	0.012
Min variance	3.2%	5.0%	0.32	0.44	-13.4%	0.24	4.5%	3.9	-0.13	0.330
Max Sharpe	8.0%	7.9%	0.80	1.28	-20.3%	0.39	13.9%	3.1	+0.35	0.034
Risk parity	4.4%	6.8%	0.42	0.61	-17.4%	0.25	2.6%	8.8	-0.03	0.641
Black-Litterman	8.0%	9.0%	0.71	1.09	-15.7%	0.51	33.0%	4.3	+0.26	0.106

Table 2. Out-of-sample performance net of 10 bp one-way costs, May 2012 – September 2026 (173 months). p-values from a 5,000-draw stationary bootstrap of the Sharpe difference vs. 1/N.

60/40 wins. The two-asset benchmark has the highest net Sharpe ratio (0.83) and the highest return (9.2% a year). Its advantage over 1/N is the only one that is significant at 5% and also survives every robustness check below. This is not an optimization result: it reflects the exceptional run of US large caps after 2012.

Naive maximum Sharpe beats 1/N here, unlike in DeMiguel et al. Its net Sharpe ratio is 0.80 against 0.45 for 1/N (p = 0.034). Two features of this experiment favour it. First, the 40% cap and long-only constraint act as strong shrinkage (Jagannathan and Ma, 2003): the optimizer cannot take the extreme positions that ruin unconstrained mean-variance portfolios. Second, momentum in asset-class returns was persistent: the optimizer held SPY and Treasuries through the 2010s and moved into gold and commodities after 2022, just before gold's rally (Figure 3). It pays for this with average monthly turnover of 13.9%, about 7 times that of 1/N, which costs it 0.02 of Sharpe ratio at 10 bp.

Risk-based methods do not beat 1/N. Minimum variance halves volatility (5.0%) and has the smallest drawdown, but its bond-heavy portfolio (on average 65% in IEF and TIP) earned little and was hit by 2022. Its Sharpe ratio (0.32) is the lowest of the six. Risk parity is close to 1/N in Sharpe (0.42, p = 0.64) with lower volatility.

Black-Litterman earns a Sharpe ratio of 0.71 with the second-smallest drawdown of the six (-15.7%, after minimum variance), but the monthly momentum view flips often and turnover is 33.0% a month, the highest of all. Its advantage over 1/N is not significant (p = 0.11).

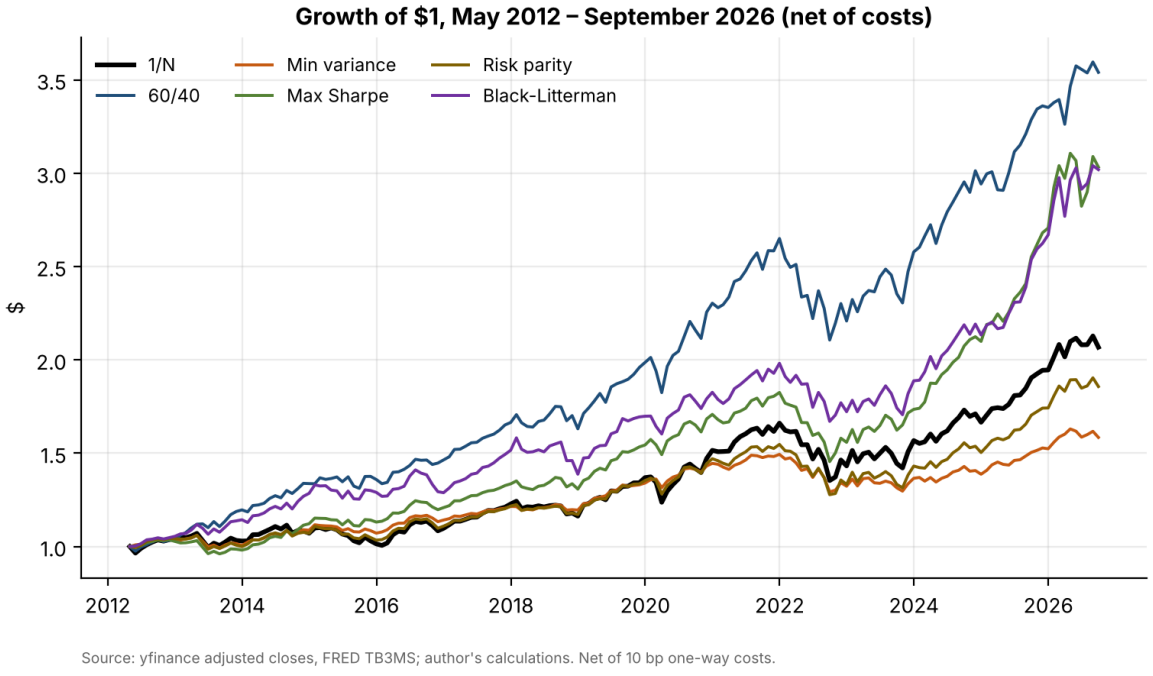
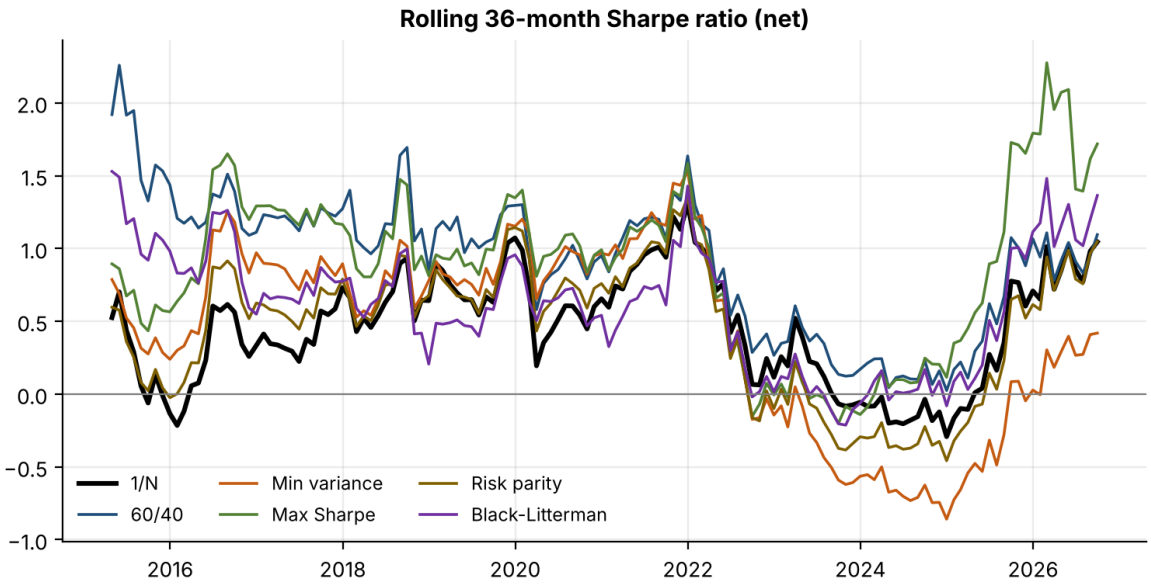
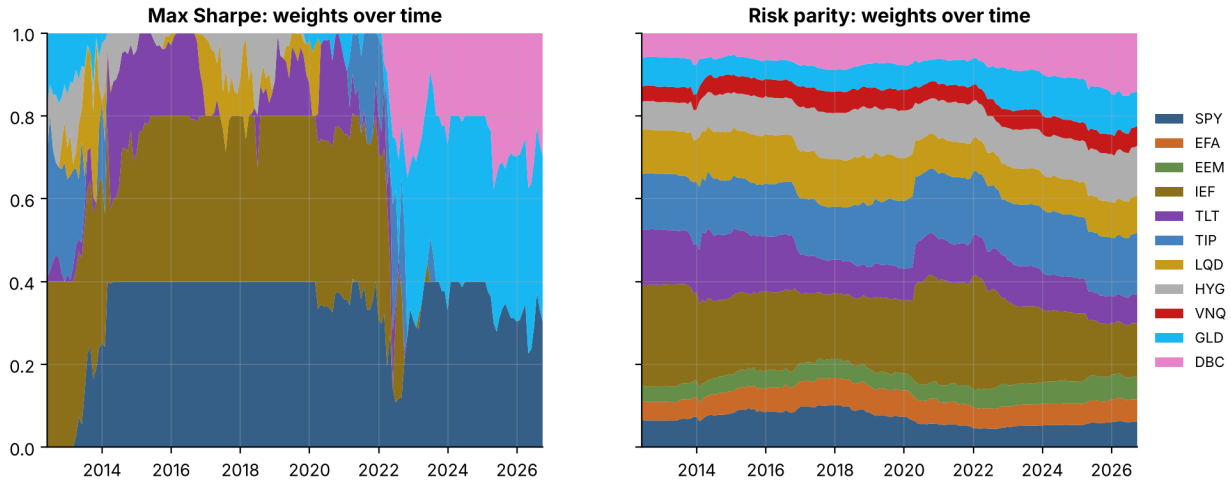


Figure 1. Growth of $1 invested in May 2012, net of 10 bp one-way costs.



Source: yfinance adjusted closes, FRED TB3MS; author's calculations. Net of 10 bp one-way costs.

Figure 2. Rolling 36-month Sharpe ratio of net excess returns. Rankings are unstable: every strategy except 60/40 spends long stretches below 1/N.



Source: author's calculations. 60-month window, 40% cap, long-only.

Figure 3. Monthly target weights of the maximum Sharpe and risk parity portfolios. Maximum Sharpe concentrates in 3–4 ETFs and hits the 40% cap most months; risk parity keeps every ETF and changes slowly.

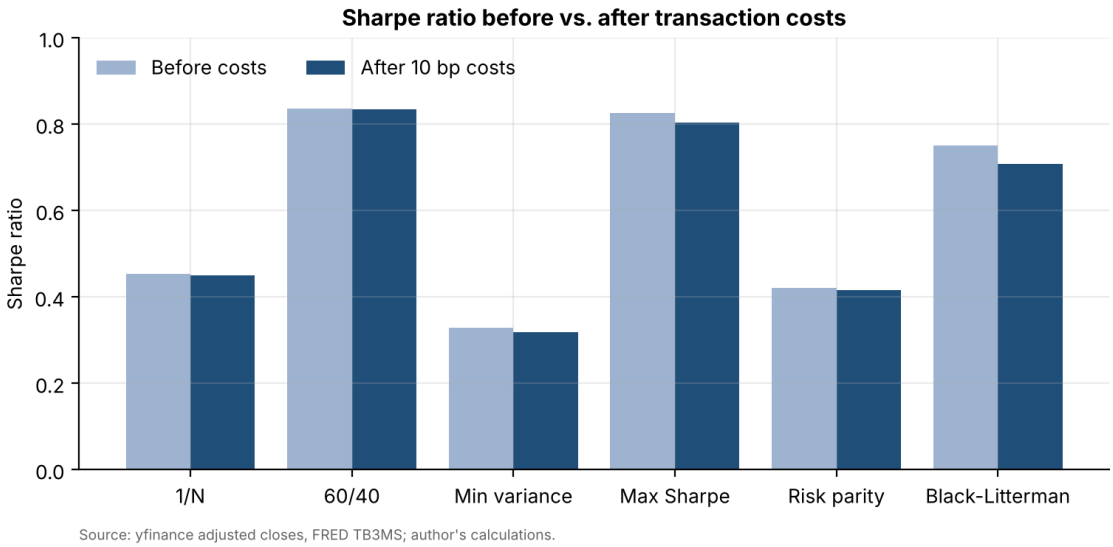


Figure 4. Sharpe ratio before and after 10 bp one-way costs. Costs matter most for the two strategies with high turnover, maximum Sharpe and Black-Litterman.

5. Robustness

Strategy	Sharpe 60m, 10bp	p vs 1/N	Sharpe 36m, 10bp	p vs 1/N	Turnover 36m	Sharpe 60m, 25bp	p vs 1/N
1/N	0.45	–	0.45	–	2.1%	0.45	–
60/40	0.83	0.012	0.83	0.012	1.6%	0.83	0.011
Min variance	0.32	0.330	0.32	0.289	6.3%	0.30	0.289
Max Sharpe	0.80	0.034	0.77	0.102	16.8%	0.77	0.054
Risk parity	0.42	0.641	0.41	0.635	3.4%	0.41	0.615
Black-Litterman	0.71	0.106	0.72	0.094	38.8%	0.64	0.216

Table 3. Robustness. The 36-month window is evaluated over the same May 2012 – September 2026 months as the base case, so only the estimation window changes.

With a 36-month window, estimation error rises and turnover of the maximum Sharpe portfolio increases from 13.9% to 16.8% a month; its Sharpe ratio falls to 0.77 and the difference from 1/N is no longer significant ($p = 0.102$). At 25 bp costs it falls to 0.77 ($p = 0.054$) and Black-Litterman falls to 0.64. The 60/40 result is unchanged in both cases.

Strategy	May 2012 - Dec 2023 Sharpe	May 2012 - Dec 2023 p vs 1/N	Jan 2024 - Sep 2026 Sharpe	Jan 2024 - Sep 2026 p vs 1/N
1/N	0.37	–	0.88	–
60/40	0.81	0.007	0.95	0.878
Min variance	0.33	0.766	0.27	0.100
Max Sharpe	0.55	0.229	1.62	0.165
Risk parity	0.32	0.509	0.89	0.962
Black-Litterman	0.55	0.300	1.28	0.228

Table 4. Subperiods. Sharpe ratios and bootstrap p-values vs. 1/N, before and after 2024.

Splitting the sample shows how fragile the optimizer's win is. Before 2024, maximum Sharpe's lead over 1/N is not significant ($p = 0.23$). Its full-sample significance comes from combining a modest, noisy lead in 2012–2023 with a very strong 2024–2026 (Sharpe 1.62), when it held up to 40% gold. Only 60/40's lead is significant in the long first subperiod.

6. Conclusion

In this 11-ETF universe over 2012–2026, the answer to DeMiguel, Garlappi and Uppal's question is mixed. None of the risk-based optimizers beats 1/N. The naive maximum-Sharpe optimizer does beat 1/N after 10 bp costs, by 0.35 of Sharpe ratio, but the result is not robust to a shorter estimation window, higher costs or

splitting the sample, so I would not rely on it. The strongest result is that a plain 60/40 portfolio beat everything, which says more about the period (US large caps returned 14.5% a year while Treasuries, TIPS and commodities returned -0.1% to 2.1%) than about portfolio construction. For an allocator the practical lessons are: constraints are a form of shrinkage and make optimizers far less fragile; turnover costs must be in the backtest; and an equal-weight benchmark is only meaningful if the universe itself is chosen sensibly.

Limitations

- One 14-year out-of-sample path; the 2012–2026 period favoured US large caps, which flatters 60/40.
- The ETF universe was chosen in 2026 with hindsight; all 11 funds survived.
- Costs are a flat 10 bp one-way; there are no taxes, bid-ask spreads beyond that, or market impact.
- Yahoo Finance adjusted prices assume dividends are reinvested at the close on the ex-date.

Appendix

Year	1/N	60/40	Min variance	Max Sharpe	Risk parity	Black-Litterman
2012 (May–Dec)	4.6%	3.5%	3.8%	2.8%	4.2%	5.5%
2013	-1.5%	15.6%	-3.0%	-4.5%	-4.1%	8.4%
2014	4.1%	11.8%	8.1%	14.4%	7.2%	12.4%
2015	-5.3%	1.6%	-1.5%	0.8%	-3.4%	0.4%
2016	8.2%	7.7%	6.6%	6.9%	7.3%	-0.1%
2017	11.9%	13.7%	6.4%	10.5%	9.7%	17.6%
2018	-5.3%	-2.0%	-1.6%	-2.1%	-3.8%	-8.5%
2019	17.7%	21.8%	12.1%	18.0%	15.6%	22.4%
2020	10.7%	16.0%	7.9%	10.6%	8.6%	7.5%
2021	9.6%	15.0%	3.3%	6.9%	5.2%	8.5%
2022	-13.8%	-16.7%	-11.4%	-14.6%	-13.8%	-13.1%
2023	9.5%	16.8%	3.2%	11.3%	7.4%	9.6%
2024	6.3%	14.1%	1.6%	21.0%	5.1%	13.0%
2025	16.8%	13.9%	9.8%	28.9%	15.9%	25.1%
2026 (Jan–Sep)	6.3%	5.6%	4.0%	12.0%	6.6%	13.1%

Table A1. Calendar-year net returns by strategy.

ETF	Min variance	Max Sharpe	Risk parity	Black-Litterman	BL prior
SPY	6.5%	33.2%	6.9%	33.4%	35.0%
EFA	0.2%	0.0%	5.5%	11.6%	15.0%
EEM	0.0%	0.0%	4.5%	4.7%	5.0%
IEF	34.7%	27.1%	19.3%	12.4%	15.0%
TLT	0.9%	7.7%	10.3%	8.7%	5.0%
TIP	30.3%	4.3%	14.3%	2.8%	5.0%
LQD	7.6%	3.4%	10.0%	5.0%	10.0%
HYG	11.7%	3.2%	9.7%	4.9%	5.0%
VNQ	0.0%	0.0%	4.4%	5.6%	2.0%
GLD	0.9%	13.1%	6.9%	6.8%	2.0%
DBC	7.2%	8.0%	8.2%	4.0%	1.0%
Months at 40% cap (any ETF)	58	155	0	57	0

Table A2. Average target weights over the out-of-sample period, and the number of months (out of 173) in which at least one ETF was at the 40% cap.

TLT	IEF	TIP	DBC	LQD	HYG	EEM	GLD	VNQ	EFA	SPY
-0.1%	1.0%	1.6%	2.1%	2.7%	4.4%	5.5%	6.1%	6.4%	7.9%	14.5%

Table A3. Annualized total return of each ETF over the out-of-sample period.

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